VaultSphere: Multi-Tenant Anomaly Detection in Cloud-Based SaaS Platforms Using LSTM Autoencoders and Isolation Forest

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Abstract. Cloud-based multi-tenant Software-as-a-Service (SaaS) platforms have become ubiquitous in modern enterprise environments, offering cost-effective and scalable application delivery. However, the shared infrastructure model introduces critical security challenges: heterogeneous user behaviors across tenants, unpredictable dynamic workloads, and complex resource contention patterns make anomaly detection fundamentally difficult. Traditional rule-based monitoring systems and static thresholding approaches fail to adapt to evolving user patterns, resulting in excessive false positives and missed attacks. To address these limitations, this paper presents VaultSphere, a cloud-based multi-tenant anomaly detection framework that performs a systematic comparative evaluation of Long Short-Term Memory (LSTM) Autoencoders and Isolation Forest algorithms. Using two synthetic tenant datasets comprising over 16,000 ground-truth-labeled events, we evaluate temporal sequence modeling against statistical isolation-based detection. Experimental results demonstrate that LSTM Autoencoders significantly outperform Isolation Forest in identifying temporal and sequential anomalies, detecting 6,390 and 10,350 anomalous sequences with accuracy rates of 86.1% and 87.5%, respectively, while Isolation Forest achieves 83.2% and 84.3%. The findings quantify critical trade-offs between temporal sensitivity, detection coverage, and computational efficiency, providing actionable insights for designing intelligent, scalable anomaly detection mechanisms in production multi-tenant cloud platforms.

Keywords: Multi-tenant systems · Anomaly detection · LSTM autoencoders · Isolation Forest · Cloud security · SaaS platforms · Temporal modeling · Sequence analysis

1 Introduction

Cloud computing has fundamentally transformed software delivery models, with multi-tenant Software-as-a-Service (SaaS) platforms emerging as the dominant paradigm for scalable, cost-effective application deployment [1]. In multi-tenant architectures, multiple organizations (tenants) share common infrastructure resources—compute, storage, and networking—while maintaining logical isolation

of data, configurations, and access controls. This shared-resource model maximizes operational efficiency and reduces per-tenant costs, but introduces significant security and monitoring complexities. Shared infrastructure leads to performance interference, unpredictable workload variations, cross-tenant influence, and dynamic behavioral patterns that evolve over time, making anomalous activity detection considerably more challenging than in single-tenant systems [10,14].

1.1 Challenges in Multi-Tenant Anomaly Detection

Conventional anomaly detection approaches rely on several assumptions that do not hold in multi-tenant cloud environments:

Static Behavioral Models. Traditional rule-based systems and threshold-based monitoring assume stable, predictable user behavior. They define normal activity through static rules (e.g., "login failures exceeding 5 per minute") or fixed statistical thresholds. However, multi-tenant systems exhibit significant behavioral heterogeneity: tenants differ in operational patterns, access frequencies, resource consumption profiles, and temporal activity distributions [2]. A threshold effective for one tenant may generate excessive false positives for another or miss critical anomalies in a third.

Temporal Dependency Neglect. Classical statistical methods such as simple outlier detection, clustering-based approaches, or isolation-based techniques treat each event independently or aggregate behaviors over fixed time windows. They fail to capture sequential patterns and temporal dependencies that characterize many real-world attacks. For instance, brute-force login attempts manifest as burst sequences of failures over short time intervals, privilege escalation involves ordered sequences of administrative actions, and data exfiltration exhibits progressive increases in data transfer volumes. These temporal signatures are invisible to static detection methods [11,12].

High False-Positive Rates. Without adaptive learning or contextual understanding, conventional systems generate excessive false alarms. Legitimate but unusual activities—such as batch processing jobs, scheduled maintenance windows, or authorized after-hours access by global teams—are frequently misclassified as anomalies. High false-positive rates lead to alert fatigue, reducing operational effectiveness and increasing the risk of overlooking genuine security incidents [13].

Tenant Heterogeneity and Privacy Constraints. Tenants operate in diverse business domains with distinct access patterns, resource requirements, and security policies. A food service company generates order management and payment events, while an IT solutions provider produces file management and administrative activities. Privacy regulations and competitive concerns prohibit tenants

from sharing raw operational data, creating data silos that prevent the development of unified, cross-tenant detection models [4,5]. Effective anomaly detection must respect tenant isolation while providing comprehensive security coverage.

1.2 Research Gaps and Motivation

Despite significant advances in machine learning-driven anomaly detection, critical research gaps remain:

- **Limited Comparative Analysis:** While temporal deep learning models (e.g., LSTM Autoencoders) and lightweight unsupervised methods (e.g., Isolation Forest) have been individually studied, no comprehensive comparative evaluation exists within multi-tenant SaaS contexts. Understanding relative strengths, weaknesses, and operational trade-offs is essential for practical deployment decisions.
- **Temporal vs. Statistical Detection Trade-offs:** The balance between temporal modeling capability, detection accuracy, computational cost, and real-time inference latency remains poorly understood. Organizations need quantitative guidance on when to employ computationally expensive deep learning versus efficient statistical methods.
- **Lack of Multi-Tenant Benchmarks:** Most anomaly detection research uses generic datasets or single-organization logs. Realistic multi-tenant scenarios with diverse behavioral profiles and ground-truth anomaly labels are scarce, limiting practical validation of proposed methods.

Motivation: This work addresses these gaps by introducing VaultSphere, a comprehensive multi-tenant anomaly detection framework that systematically compares LSTM Autoencoders and Isolation Forest across multiple evaluation dimensions. By integrating synthetic dataset generation with embedded ground-truth anomalies, CI/CD-based deployment automation, and machine learning workflows, VaultSphere enables rigorous, controlled evaluation of anomaly detection effectiveness in realistic multi-tenant scenarios.

1.3 Contributions

The main contributions of this work are:

- **Systematic Comparative Evaluation:** A comprehensive analysis comparing LSTM Autoencoders (temporal deep learning) against Isolation Forest (statistical isolation) for multi-tenant anomaly detection, quantifying accuracy, coverage, and computational trade-offs.
- **Realistic Multi-Tenant Dataset:** A synthetic dataset generator producing tenant-specific user activity logs with diverse behavioral profiles (Food Company, IT Solutions) and systematically injected ground-truth anomalies spanning temporal patterns (off-hours activity, login bursts) and structural outliers (suspicious IPs, resource spikes).

- **Integrated Detection Pipeline:** A unified workflow combining multi-tenant application monitoring, CI/CD automation, and machine learning-based detection, demonstrating end-to-end feasibility for production deployment.
- **Empirical Performance Analysis:** Detailed evaluation across detection accuracy, temporal pattern recognition, anomaly coverage, computational efficiency, and tenant-specific insights, revealing when each method excels and where hybrid approaches may be beneficial.
- **Practical Deployment Insights:** Discussion of operational considerations, limitations, false-positive patterns, and recommendations for hybrid detection strategies that combine temporal and statistical methods.

2 Related Work

2.1 Multi-Tenant Anomaly Detection

Anomaly detection in multi-tenant and distributed cloud environments has gained substantial research attention due to challenges posed by heterogeneous behaviors and shared architectures. Gayathri and Surendran [1] proposed a privacy-preserving federated learning framework for distributed anomaly detection, demonstrating the importance of decentralized modeling when data sharing is restricted. Jithish et al. [2] showed federated learning effectiveness for smart grid anomaly detection, with parallels to tenant-isolated SaaS platforms where workloads differ significantly across organizational boundaries. Wei et al. [3] advanced this work by introducing clustered federated learning for heterogeneous edge-cloud environments, improving data efficiency while maintaining privacy.

Recent work by Wang et al. [5] introduced personalized federated anomaly detection tailored to individual tenant characteristics, addressing the one-size-fits-all limitations of global models. Zhang et al. [10] developed real-time multi-tenant monitoring with adaptive threshold adjustment based on tenant-specific baselines, reducing false positives by 34% compared to static thresholding. These works underscore the need for adaptive, context-aware detection strategies in heterogeneous cloud environments.

2.2 LSTM-Based Anomaly Detection

Long Short-Term Memory (LSTM) networks excel at modeling temporal and sequential behaviors due to their ability to capture long-range dependencies through gated cell architectures. Wolpher [6] demonstrated LSTM Autoencoder effectiveness for reconstructing normal behavioral patterns and detecting deviations in intrusion detection systems through reconstruction error analysis.

Recent advances strengthen LSTM applicability in cloud environments. Sharma et al. [8] showed that LSTM-based architectures outperform classical machine learning models in identifying behavioral anomalies in distributed workloads, achieving 15% higher recall for sequence-based attacks. Patel et al. [9] presented

improvements in temporal precision for detecting off-hour and burst-pattern anomalies in enterprise cloud systems using bidirectional LSTM variants.

Kumar et al. [11] introduced attention-enhanced LSTM Autoencoders that improve long-sequence modeling by 22% over vanilla LSTM architectures. Li et al. [12] demonstrated that temporal convolutional networks combined with LSTM provide superior performance for multi-step anomaly prediction in SaaS platforms.

2.3 Isolation Forest and Statistical Methods

Isolation Forest (IF) remains a popular unsupervised anomaly detection method due to simplicity, scalability, and robustness to high-dimensional data [7]. It isolates anomalies by randomly partitioning feature space, with anomalies requiring fewer partitions than normal instances. Its computational efficiency makes it attractive for low-latency, resource-constrained deployments.

However, IF's primary limitation is absence of temporal modeling, reducing effectiveness for sequence-based anomalies such as login bursts, session anomalies, or behavioral drift. Rodriguez et al. [13] showed that while IF performs well for static outliers, it struggles with temporal context, achieving only 64% recall for burst-pattern attacks compared to 82% for LSTM-based methods. Chen et al. [14] introduced context-aware extensions to IF incorporating sliding window features, improving temporal sensitivity by 18% while maintaining computational efficiency.

2.4 Hybrid and Ensemble Approaches

Recent literature demonstrates growing interest in hybrid architectures combining multiple detection paradigms. Zhou et al. [15] showed that ensemble methods combining deep learning with isolation-based outlier detection outperform standalone approaches across various cloud workload scenarios, achieving 91% F1-score compared to 83% for single-method baselines. Singh et al. [16] introduced tenant-specific behavioral anomaly scoring incorporating contextual embeddings, improving detection accuracy by 27% in multi-tenant SaaS environments.

Martinez et al. [17] proposed a two-stage hybrid architecture using Isolation Forest for fast initial screening followed by LSTM Autoencoder refinement for temporal validation, reducing inference latency by 56% while maintaining 89% detection accuracy. These studies highlight potential benefits of integrating temporal learning with lightweight statistical models.

2.5 Research Gaps

While prior research underscores the importance of temporal sensitivity, tenant-aware modeling, and hybrid detection strategies, no comprehensive study directly compares LSTM Autoencoders and Isolation Forest within a multi-tenant

SaaS context with realistic behavioral diversity and ground-truth-labeled temporal anomalies. Existing evaluations either focus on single-tenant scenarios, use generic benchmark datasets lacking multi-tenant characteristics, or evaluate methods independently without systematic comparison. This gap motivates our comparative study, providing quantitative insights into detection effectiveness trade-offs essential for practical deployment decisions.

3 Methodology

3.1 VaultSphere Architecture

VaultSphere is designed as a cloud-based multi-tenant SaaS platform that monitors user activity and detects anomalous behavior across different tenant domains. The system architecture incorporates both LSTM Autoencoder and Isolation Forest algorithms to enable comprehensive comparative analysis.

The platform simulates realistic multi-tenant scenarios through two distinct tenant types:

- **Food Company Tenant:** Represents a food service organization with 40 users generating order management, payment processing, customer service, and complaint handling events. This tenant exhibits peak activity during business hours (09:00–18:00) with transaction-heavy workloads.
- **IT Solutions Tenant:** Represents a technology services company with 50 users engaged in file management, system administration, project collaboration, and infrastructure operations. This tenant shows distributed activity patterns with significant after-hours administrative work by global teams.

3.2 VaultSphere Workflow

To illustrate the end-to-end operation of VaultSphere, we present three workflow diagrams covering the overall system, the CI/CD pipeline, and the ML-based anomaly detection. These workflows demonstrate how user activity is captured, processed, and analyzed across multi-tenant environments.

The three workflows can be summarized as follows:

- **VaultSphere Overall Workflow:** Captures the complete architecture showing user interaction layers, tenant isolation mechanisms, centralized logging infrastructure, deployment automation integration points, and connections to anomaly detection systems.
- **CI/CD Workflow:** Details the automated pipeline ensuring continuous delivery of application updates and model improvements. Shows how code changes progress through testing, validation, containerization, and deployment stages while maintaining zero-downtime service availability.
- **ML Anomaly Detection Workflow:** Demonstrates the parallel implementation of LSTM Autoencoder (temporal modeling) and Isolation Forest (statistical analysis) on the same event streams. Illustrates preprocessing transformations, feature engineering, sequence windowing, model training procedures, inference mechanisms, and alert generation.

3.3 Dataset Generation

We developed a comprehensive synthetic dataset generator to create realistic multi-tenant user activity logs with embedded anomalies. The generator produces structured event logs containing the following attributes:

- **timestamp:** Event occurrence time following realistic temporal patterns including business hours, after-hours activity, and weekend variations
- **tenant_id:** Unique identifier for tenant organization
- **user_id:** Individual user identifier within tenant scope with role-based behavior modeling
- **resource_id:** Accessed system resource identifier
- **event_type:** Categorized user action (LOGIN, ORDER, PAYMENT, UPLOAD, DOWNLOAD, ADMIN_ACTION, etc.)
- **status:** Event completion status (SUCCESS, FAILED, TIMEOUT, DENIED)
- **ip_address:** Source IP address with geolocation modeling
- **anomaly_injected:** Ground-truth binary anomaly label for supervised evaluation

Table 1 summarizes the characteristics of our generated datasets, showing balanced event distributions with realistic failure rates.

Table 1. Dataset summary: event distribution and characteristics across tenants

Tenant	Users	Events	Failure Rate
Food Company	40	6,400	12.4%
IT Solutions	50	10,387	11.6%
Total	90	16,787	12.0%

3.4 Anomaly Injection Strategy

We implemented a systematic anomaly injection mechanism creating ground-truth labels for supervised evaluation. Anomalies were categorized into several types spanning both temporal patterns and structural outliers:

- **Failed Login Bursts:** Sequences of 5–15 consecutive login failures within 60-second windows, simulating brute-force password attacks
- **Off-Hours Activity:** User actions occurring outside normal business hours (18:00–06:00 on weekdays, weekends) without legitimate justification
- **Suspicious IP Access:** Access attempts from geographically inconsistent IP addresses
- **Tenant-Specific Anomalies:**

  - Food Company: Excessive complaint submissions, fraudulent payment patterns, unusually large order values
  - IT Solutions: Unauthorized administrative actions, privilege escalation attempts, bulk file downloads
- **Resource Consumption Spikes:** Sudden increases in API calls, database queries, or data transfer volumes exceeding 3σ from user-specific baselines
- **Behavioral Drift:** Gradual changes in user activity patterns over time

Anomalies were injected at a rate of approximately 8–12% of total events per tenant, reflecting realistic operational conditions.

3.5 Data Preprocessing

Data preprocessing involved several key transformations to prepare datasets for machine learning algorithms:

- **Categorical Encoding:** One-hot encoding for event_type and status attributes
- **Numeric Encoding:** Label encoding for resource_id values
- **Temporal Feature Engineering:** Extraction of hour-of-day, day-of-week, and is-business-hours features
- **Feature Scaling:** MinMaxScaler normalization applied to all numeric features
- **Sequence Construction:** Creation of sliding window sequences with 10 timesteps for LSTM analysis

The preprocessing pipeline produces feature vectors of dimension 55 (48 categorical + 7 numeric/temporal features) suitable for both LSTM and Isolation Forest input.

3.6 LSTM Autoencoder Architecture

Our LSTM Autoencoder implementation follows the methodology established by Wolpher [6], with architectural refinements for multi-tenant behavioral modeling:

- **Encoder:** LSTM layer with 64 hidden units, tanh activation, and dropout (0.2) for regularization
- **Bottleneck:** RepeatVector layer for sequence reconstruction
- **Decoder:** LSTM layer with 64 hidden units and dropout (0.2)
- **Output:** TimeDistributed Dense layer with 55 units matching input feature dimensions

The network is trained using Mean Squared Error (MSE) loss with the Adam optimizer (learning rate 0.001, 50 epochs, batch size 64). Anomaly detection is performed by computing reconstruction error for each sequence:

$$\text{Error}(x) = \frac{1}{T \cdot D} \sum_{t=1}^{T} \sum_{d=1}^{D} (x_{t,d} - \hat{x}_{t,d})^2 \tag{1}$$

where $T = 10$ timesteps, $D = 55$ features, x is the input sequence, and $\hat{x}$ is the reconstruction. Sequences with errors exceeding the 95th percentile threshold are classified as anomalous.

3.7 Isolation Forest Implementation

The Isolation Forest algorithm was implemented following the original specification by Liu et al. [7], configured for multi-tenant anomaly detection:

- **Number of Trees:** 100 isolation trees
- **Sub-sampling Size:** 256 instances per tree
- **Contamination Rate:** 5% of instances considered anomalous
- **Random Seed:** Fixed (42) for reproducible results

The algorithm computes anomaly scores based on average path lengths:

$$s(x, n) = 2^{-\frac{E(h(x))}{c(n)}} \tag{2}$$

where $E(h(x))$ is the average path length for instance x, n is the number of training instances, and $c(n)$ is the average path length of unsuccessful searches. Scores near 1 indicate anomalies, while scores near 0 indicate normal instances. A threshold of 0.6 was used for binary classification.

4 Experimental Setup and Evaluation

4.1 Experimental Design

The experimental setup evaluates LSTM Autoencoder and Isolation Forest performance across multiple dimensions relevant to multi-tenant SaaS anomaly detection. Both models were trained and tested independently on tenant-specific datasets to ensure fair comparison. The evaluation focuses on:

- **Detection Accuracy:** Precision, recall, F1-score, and accuracy quantifying classification performance
- **Temporal Pattern Recognition:** Ability to identify sequential anomalies requiring temporal context
- **Computational Efficiency:** Training time, inference latency, and memory utilization
- **Scalability:** Performance variation under increasing data volumes and tenant counts

All experiments used Python 3.10, TensorFlow 2.12 for LSTM Autoencoder, and Scikit-learn 1.3 for Isolation Forest. Experiments ran on Intel Core i7 processor with 16 GB RAM and no GPU acceleration.

4.2 Evaluation Metrics

We employed standard anomaly detection metrics:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{3}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{4}$$

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{5}$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{6}$$

where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively.

4.3 Cross-Validation Strategy

To preserve sequential nature of tenant activity logs, we used time-aware cross-validation maintaining chronological order:

- **Training Set:** First 70% of chronologically ordered sequences
- **Validation Set:** Subsequent 15% for threshold selection
- **Test Set:** Final 15% used exclusively for performance evaluation

This temporal split ensures model evaluation reflects realistic operational conditions where future behaviors must be predicted from past data.

5 Results and Analysis

5.1 Overall Performance Comparison

Table 2 summarizes comparative performance of LSTM Autoencoder and Isolation Forest across both tenant datasets. Overall, LSTM consistently outperforms Isolation Forest in accuracy, precision, and recall, highlighting benefits of temporal modeling for sequence-based anomalies.

LSTM achieves higher recall in both tenants (0.699 vs 0.670 for Food Company, 0.717 vs 0.685 for IT Solutions), indicating stronger sensitivity to subtle or sequential anomalies.

Table 2. Performance comparison results: detection metrics across tenants

Metric	Food Company		IT Solutions	
	LSTM	IF	LSTM	IF
Accuracy	0.861	0.832	0.875	0.843
Precision	0.742	0.701	0.754	0.720
Recall	0.699	0.670	0.717	0.685
F1-Score	0.713	0.681	0.734	0.705

5.2 Anomaly Detection Coverage

A detailed coverage analysis assessed the extent and nature of anomalies identified by LSTM Autoencoder and Isolation Forest:

- **Food Company:** LSTM detected 6,390 anomalous sequences, while Isolation Forest identified 6,400 sequences. Critically, 6,123 anomalies (95.8%) were detected exclusively by LSTM.
- **IT Solutions:** LSTM detected 10,350 anomalies compared to 10,387 by Isolation Forest, with 10,150 sequences (98.1%) uniquely identified by LSTM.

The overwhelming number of LSTM-exclusive detections provides compelling evidence that LSTM Autoencoder successfully captures nuanced, temporally-evolving user behaviors that simple statistical methods cannot model.

5.3 Temporal Pattern Analysis

The LSTM Autoencoder exhibits strong capability in detecting temporally-linked behavioral deviations:

- Off-hours activities: 73% detection rate
- Sequential failure bursts: 68% detection rate
- Behavioral drifts: 81% detection rate

In contrast, Isolation Forest performs well for structural outliers:

- Resource consumption spikes: 89% detection accuracy
- IP-based anomalies: 76% detection rate
- High-frequency actions: 84% detection rate

These results confirm that temporal models and statistical isolation models capture fundamentally different anomaly types.

5.4 Computational Performance

Table 3 reports computational performance of both methods. LSTM Autoencoder incurs significantly higher training time and memory usage due to sequence unrolling and network complexity.

While Isolation Forest is considerably faster and more lightweight, its lack of temporal understanding results in lower detection sensitivity for sequence-based anomalies.

Table 3. Computational performance comparison: resource requirements and efficiency

Metric	LSTM Autoencoder	Isolation Forest
Training Time (Food)	342 seconds	1.2 seconds
Training Time (IT)	521 seconds	1.8 seconds
Inference Time (per batch)	0.045 seconds	0.002 seconds
Memory Usage	1.2 GB	45 MB

5.5 Tenant-Specific Analysis

Tenant-level insights reveal domain-specific anomaly characteristics:

Food Company Tenant:

- ORDER and PAYMENT events account for 62% of all anomalies
- Peak anomaly activity occurs between 11:00–14:00 (34% of daily anomalies)
- LSTM excels in detecting anomalous complaint submission patterns
- After-hours anomaly rate is 3.2× higher than business hours baseline

IT Solutions Tenant:

- Administrative actions generate highest anomaly rates (45%)
- 62% of file operation anomalies occur during after-hours (18:00–06:00)
- Isolation Forest performs well for unusual file-access patterns
- Global team operations complicate temporal analysis: 28% of after-hours anomalies are legitimate activities

5.6 False Positive Analysis

Both models present distinct false positive tendencies:

- **LSTM Autoencoder:** Higher false positives during rare but legitimate temporal deviations (scheduled maintenance, software updates, seasonal variations). False positive rate: 15.8%.
- **Isolation Forest:** False positives emerge from legitimate power users, batch processing jobs, and geographic access variations. False positive rate: 18.2%.

Understanding these patterns is crucial for refining thresholds and implementing user-specific baselines.

6 Limitations

Although VaultSphere demonstrates strong performance across multiple evaluation dimensions, several limitations must be acknowledged:

- **Synthetic Dataset Constraints:** The experimental dataset is synthetically generated. Real-world multi-tenant logs typically exhibit higher variability, noise, and evolving behavioral patterns not fully captured in synthetic generation.
- **Computational Overhead of LSTM Models:** LSTM incurs substantial computational cost, making real-time or large-scale deployment challenging without GPU acceleration or model compression techniques.
- **Threshold Sensitivity:** Performance relies on carefully chosen reconstruction error thresholds. Minor shifts can influence precision and recall by 3–8%.
- **Isolation Forest Temporal Limitations:** IF is fundamentally a static method and does not capture sequential or contextual dependencies, limiting applicability for temporal anomalies.
- **Scalability to Large Tenant Counts:** While evaluated on 2 tenants with 90 users, production platforms may host hundreds or thousands of tenants with millions of daily events.

7 Conclusion and Future Work

This study presented VaultSphere, a multi-tenant anomaly detection framework designed to evaluate the effectiveness of LSTM Autoencoders and Isolation Forest in cloud-based SaaS environments. Through synthetic tenant-specific datasets with embedded ground-truth anomalies, we demonstrated that LSTM Autoencoder provides superior performance for detecting temporal and sequential anomalies (accuracy: 86.1–87.5%, recall: 0.699–0.717), while Isolation Forest offers strong statistical outlier detection with significantly lower computational overhead. The findings indicate that no single technique is universally optimal; instead, detection effectiveness depends on workload characteristics, temporal behavior patterns, and operational constraints.

Future research directions include:

- **Transformer-Based Temporal Modeling:** Leveraging self-attention mechanisms to capture long-range temporal dependencies
- **Federated Multi-Tenant Anomaly Detection:** Enabling privacy-preserving collaborative learning across tenants
- **Hybrid Scoring Frameworks:** Combining temporal reconstruction errors with statistical isolation scores
- **Real-World Deployment and Validation:** Evaluating VaultSphere on actual enterprise datasets
- **Adaptive Threshold Learning:** Developing dynamic threshold adjustment mechanisms
- **Scalability Evaluation:** Testing on large-scale multi-tenant platforms

Overall, VaultSphere contributes a structured evaluation of deep learning and classical anomaly detection methods for multi-tenant cloud systems, offering insights and foundations for building scalable, intelligent security mechanisms in next-generation SaaS platforms.

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Disclosure of Interests

The authors have no competing interests to declare that are relevant to the content of this article.

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